1 Introduction and inequalities

Week 1 · Cheat sheet

Interval	**x < -4**	**-4 < x < 3**	**x > 3**
Test value	x = -5	x = 0	x = 4
x + 4	**-**	**+**	**+**
x - 3	**-**	**-**	**+**
(x + 4)(x - 3)	**+**	**-**	**+**

> 0



Intervals

$[a, b] \Leftrightarrow a \le x \le b$	Both ends included
$(a, b) \Leftrightarrow a < x < b$	Both ends excluded
$[a, \infty) \Leftrightarrow x \ge a$	∞ always takes a parenthesis
$A \cup B$ / $A \cap B$	"or" (either) / "and" (overlap)
$\mathbb{R} = (-\infty, \infty)$, $\varnothing$	All reals / no solution

Inequality rules

$a < b \Rightarrow a + c < b + c$	Adding keeps the sign
$c > 0$: $a < b \Rightarrow ac < bc$	Positive factor keeps it
$c < 0$: $a < b \Rightarrow ac > bc$	Negative factor flips it
$P(x)/Q(x) \ge 0$	Never multiply by $Q(x)$: use a sign chart

Equations

$ax + b = 0 \rightarrow x = -b/a$	Linear: one root
$x = (-b \pm \sqrt{(b^2 - 4ac)}) / 2a$	Quadratic formula
$\Delta = b^2 - 4ac$	> 0: two roots, = 0: one, < 0: none
$A \cdot B = 0 \Leftrightarrow A = 0$ or $B = 0$	Zero product rule
p/q: $p \mid a_0$, $q \mid a_n$	Rational root candidates

Absolute value

$\lvert x - a \rvert$	Distance from x to a
$\lvert x - a \rvert < r \Leftrightarrow a - r < x < a + r$	One interval
$\lvert x - a \rvert > r \Leftrightarrow x < a - r$ or $x > a + r$	Two pieces

Sign chart in five steps	
1. … ≥ 0	Move everything to one side
2. (x - r_1)(x - r_2)…	Factor
3. r_1, r_2, …	Critical points: zeros and undefined points
4. one test value per interval	The sign is constant between them
5. ∪ of the right intervals	Endpoints: [ ] for ≤ ≥, ( ) for < > and denominators

When something goes wrong

Symptom	Likely cause	Fix
Answer is the mirror of the right one	Divided by a negative without flipping	Test one value from your answer in the original
A point makes the expression undefined	A zero of the denominator was included	Always use ( ) at denominator zeros
$x^2 < 9$ solved as $x < 3$	Took a square root like an equation	Factor (x - 3)(x + 3) and use a sign chart: (-3, 3)